

Jonathan Ma

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EDUCATION

Johns Hopkins University

Baltimore, MD

Master of Science in Engineering, Applied Mathematics and Statistics

Expected Dec 2026

Coursework: Statistical and AI Methods in Clinical Data Science, Nonlinear Optimization, Time Series Analysis

Rutgers University

Camden, NJ

Bachelor of Science in Finance and Economics, Statistics Minor

May 2025

GPA: 3.929 / 4.0

Honors: Economics Department Award, Summa Cum Laude, Omicron Delta Epsilon

WORK & LEADERSHIP EXPERIENCE

University of Utah, Sharon Eccles Steele Center for Translational Medicine

Salt Lake City, UT

Affiliate Researcher

Jun 2026 – Aug 2026

- Developed computational pipelines for quantifying hypertransmission biomarkers in retinal OCT images by integrating Heidelberg Eye Explorer (HEYEX) segmentations with logistic regression-based feature weighting and sensitivity analyses for age-related macular degeneration research.
- Designed and evaluated image analysis and deep learning pipelines, including ResNet50 transfer learning, Grad-CAM explainability, and bounding-box localization methods on 24,000+ OCT images, achieving approximately 93% classification accuracy.
- Collaborated with ophthalmologists, clinicians, and biostatisticians to develop HIPAA-compliant imaging workflows, clinician-assisted annotation software, and interpretable biomarkers for disease progression modeling.

Rutgers University

Camden, NJ

Research Assistant

Oct 2023 – Jul 2025

- Built an end-to-end financial text analytics pipeline by collecting and preprocessing over 7,500 earnings call transcripts using Python and R, producing a structured corpus for downstream NLP and forecasting analyses.
- Developed neural time-series forecasting models using N-BEATS and N-HiTS, achieving an 18–25% reduction in RMSE compared with ARIMA and LSTM baselines through feature engineering, hyperparameter optimization, and rolling-window validation.
- Constructed and validated a database of 23,500+ equity analysts, developing an automated entity-resolution pipeline that reduced 3,750 ambiguous records to 54 verified unresolved cases through Selenium-assisted cross-source verification.

PROJECTS & PUBLICATIONS

innovcal: Calibration-Aware Diffusion Forecasting (Upcoming)

- Developed calibration-aware diffusion models for probabilistic forecasting by integrating CRPS-based objectives, differentiable calibration losses, and score-based generative modeling with VAR and DeepAR forecasting systems.
- Evaluated calibration using PIT, CRPS, Expected Calibration Error, energy scores, and robustness analyses under controlled heavy-tailed, mixture, and heteroskedastic innovation processes.

Bayesian Hierarchical Modeling for Malignant Cancer Diagnosis (Link)

- Developed a three-level Bayesian hierarchical logistic regression model to quantify patient-, tumor-, and region-level determinants of late-stage cancer diagnosis using SEER data.
- Performed full Bayesian inference using NUTS, posterior predictive validation, convergence diagnostics, and Gaussian mixture modeling to identify clinically meaningful patient risk subgroups while quantifying uncertainty across hierarchical levels.

TECHNICAL SKILLS

Programming: Python, R, SQL, Julia, C++, MATLAB, JavaScript, Git, STATA, SAS, ImageJ, HEYEX

Certifications: CITI Good Clinical Practice (2026), CFI FMVA (2024), Bloomberg Market Concepts (2022)